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B.Tech Project Presentation

StudyBuddy AI

AI-Powered Academic Learning Assistant for B.Tech Students

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B.Tech · Computer Science and Artificial Intelligence & Machine Learning · Individual project

Outline

Roadmap of this presentation

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Outline	**Introduction**	**Objectives**	**Methodology Used**	**Implementation**	**Conclusion**	**References**
This roadmap	Problem and motivation	What the prototype aims to do	Architecture, workflow, technologies	Dashboards, data, AI, notes	Limitations, scope, summary	Documentation and artifacts

Slides 3–8 follow the sections above in order

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Introduction

The academic problem StudyBuddy AI addresses

Many subjects, many semesters

Engineering students handle subjects, semester-wise material, assessments and revision.

Scattered information

Learning resources and performance details are hard to keep in one place.

Need for quick help

Students want fast explanations and a structured way to review weak topics.

Teachers need an overview

A consolidated view of marks and weak topics supports revision planning.

Fragmented academic workflow → **StudyBuddy AI** → **More organized learning workflow**

StudyBuddy AI is a prototype that brings learning and academic-support functions into one interface. No improvement in grades or learning outcomes has been measured.

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Objectives

What the prototype sets out to provide

Unified role-based interface

Student and Teacher dashboards in one Streamlit app.

Implemented

Semester and subject structure

8 AKTU CSE semesters with fixed subject lists.

Implemented

Visual analytics

Plotly charts for trends and class performance.

Demo data

Teacher tools

Marks editor, assignment list, weak-unit view.

Demo data

AI-assisted explanations

Chat tutor via OpenAI API when a key is entered.

Implemented

PYQ and notes

PYQ upload and a persistent local notes library.

Partial

Study planning

Checklist, progress bar and month-based tips.

Implemented

Status key

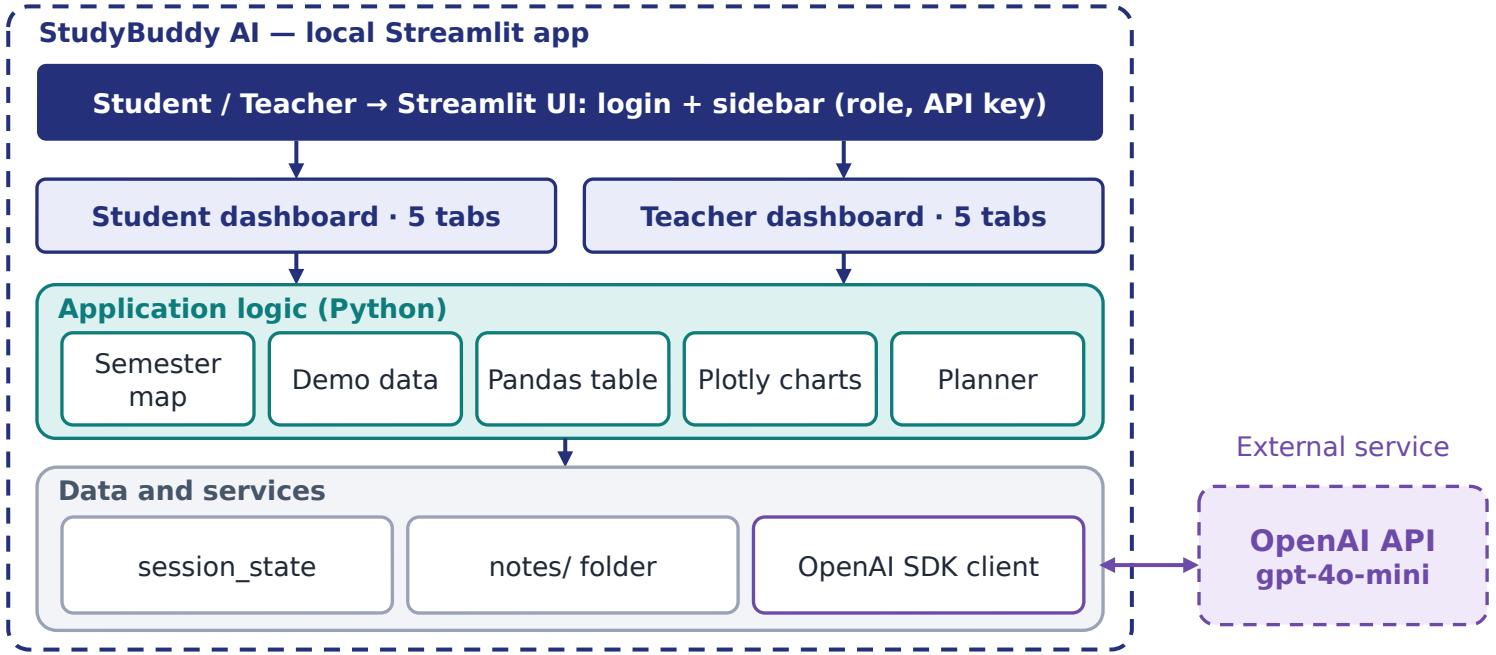
Implemented	works as coded
Demo data	random / sample values
Partial	some parts missing

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Methodology Used

Architecture, workflow and technologies

System architecture



Application workflow

1. Open app and log in (any non-empty values pass)
2. Choose role: Student or Teacher
3. Optional: enter OpenAI API key in sidebar
4. Tabs prepare data: fixed lists, random demo values
5. Render charts, planner, uploads; AI only on request
6. Show results; warning or error if key or API fails

Python app logic

Streamlit web interface

Pandas marks table

Plotly charts

OpenAI SDK AI requests

pathlib notes storage

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Implementation

Dashboards, data, AI Companion and notes

Student dashboard

Demo data

- Sidebar: user, logout, optional API key
- 5 tabs: Learning Hub, Analytics, Planner, AI Companion, PYQ & Notes
- Analytics are random demo values; alerts are static samples

Teacher dashboard

Demo data

- 5 tabs: Performance, Assignments, Marks Record, Topic Difficulty, Student Progress
- Editable marks table kept for the session
- “AI Insight” text is fixed or rule-based

Data and analytics

No CSV

Random / fixed data → Pandas table → Plotly chart

Weak unit: any student scoring below 60 in its mapped column

Demo data — not real student results

AI Companion

Needs API key

Student question → Streamlit chat box → Python + OpenAI SDK → gpt-4o-mini (external) → Answer in chat

System prompt + last 10 messages are sent. No key: warning, no request. Errors are shown. Needs internet and billing; subject and notes are not sent.

PYQ and notes

Partial

PYQ	Upload PDF/TXT	Session memory	TXT preview	AI analysis
Notes	Upload PDF/TXT/MD	Save to notes/	Preview	Download

Not implemented: PDF text extraction, note search

Conclusion

Summary, limitations and future scope

Summary

- One interface for learning, planning, analytics, AI chat and notes
- Student and teacher dashboards
- Plotly analytics (demo data)
- AI explanations via an external OpenAI model
- No measured learning outcomes are claimed

Current limitations

- Prototype login accepts any non-empty credentials
- Random demo data, not real records
- No database; session data is lost (only notes persist)
- AI needs internet, an API key and billing
- No PDF text extraction

Future scope (planned)

- Verified AKTU previous-year paper collection
- Database, accounts and secure role-based access
- Answers retrieved from uploaded notes
- PDF text extraction and search
- Cloud storage, multi-user deployment, revision tips from validated data

Now: Streamlit prototype with demo data → **Next: real data, database, secure login** → **Later: notes-based answers, cloud, multi-user**

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References

Documentation and project artifacts

Python documentation
https://docs.python.org/3/

Streamlit documentation
https://docs.streamlit.io/

Pandas documentation
https://pandas.pydata.org/docs/

Plotly Python documentation
https://plotly.com/python/

OpenAI API documentation (Chat Completions)
https://developers.openai.com/api/docs/

Project artifacts (not external publications)

- studybuddy.py — Streamlit source code
- notes/ — local folder created at runtime for saved notes

No external dataset

The prototype uses no external dataset, so no dataset source is cited.

Thank You · Anshika Pandey · University Roll No. 2400101530028 · StudyBuddy AI